

MD NAZRUL ISLAM

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RESEARCH INTERESTS

Vision-language models · Evaluation validity · Explanation faithfulness · Low-resource NLP robustness

EDUCATION

Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh

Jan 2018 – Feb 2023

- B.Sc. in Computer Science & Engineering. CGPA 3.68 / 4.00, ranked 25th of 127 in the graduating class (top 20%).
- **IELTS Academic** (Oct 2024): 7.5 overall — L 7.5, R 8.0, W 6.5, S 7.0.

PUBLICATIONS

Preprints and papers under review (* equal contribution; my role stated for each paper)

1. Md Asiful Islam Miah*, **Md Nazrul Islam***, S M Taslim Uddin Raju, Hamdi Altaheri. “GeoToken: Differentiable 3D Voxel Tokenization for Geometric Spatial Reasoning in VLMs.” *Under review*. [Paper](#) · Code available on request (double-blind review).
 - A 1B VLM whose tokenizer lifts image patches into metric 3D voxel tokens: at 3.6 GB and with no region proposals it reaches 81.33% on SpatialRGPT-Bench, within 10.5 of SpatialRGPT-7B.
 - **Role**: implemented the spatially aware projection encoder; built the training pipeline (Kaggle compute, versioned checkpointing on the Hugging Face Hub).
2. **Md Nazrul Islam**, Md Habibullah Belali. “Low Scores Without a Perceptual Deficit: Separating Answer-Position Bias from Perception on MMVP.” *Under review*. [Paper](#) · [Code](#)
 - MMVP’s below-chance scores track answer-position bias, not perception: one reversed-options control puts perceptual accuracy at 68–73 across models whose published scores span 6.0 to 33.3.
 - **Role**: problem formulation, the reversed-options control, statistical design, analysis and writing.
3. **Md Nazrul Islam**. “Character Noise Makes Bangla NER Miss Entities, Not Mislabel Them.” *Under review, ACL Rolling Review (October 2026 cycle)*. [Paper](#) · Code available on request (double-blind review).
 - With the per-character noise rate held fixed across scopes, damage per corrupted token is 9.3× higher on entity tokens than on context tokens and 86% of the loss is missed entities; character augmentation recovers up to 14.3 F1, FGSM nothing.
 - **Role**: sole author — problem formulation, controlled perturbation design, experiments, analysis and writing.

Peer-reviewed journal articles

1. Ovishake Sen, Mohtasim Fuad, **Md Nazrul Islam**, Jakaria Rabbi, Mehedi Masud, Md Kamrul Hasan, Md Abdul Awal, Awal Ahmed Fime, Md Tahmid Hasan Fuad, Delowar Sikder, Md Akil Raihan Iftee. “Bangla Natural Language Processing: A Comprehensive Analysis of Classical, Machine Learning, and Deep Learning-Based Methods.” *IEEE Access* 10 (2022): 38999–39044. [doi:10.1109/ACCESS.2022.3165563](https://doi.org/10.1109/ACCESS.2022.3165563)
 - **Role**: co-designed the eleven-task taxonomy; wrote the deep-learning method comparisons and the low-resource challenges analysis.
2. Md Tahmid Hasan Fuad, Awal Ahmed Fime, Delowar Sikder, Md Akil Raihan Iftee, Jakaria Rabbi, Mabrook S Al-Rakhani, Abdu Gumaie, Ovishake Sen, Mohtasim Fuad, **Md Nazrul Islam**. “Recent Advances in Deep Learning Techniques for Face Recognition.” *IEEE Access* 9 (2021): 99112–99142. [doi:10.1109/ACCESS.2021.3096136](https://doi.org/10.1109/ACCESS.2021.3096136)
 - **Role**: analysis of representation learning methods, loss functions and datasets.

Peer-reviewed conference paper

1. Towsif Ahamed, **Md Nazrul Islam**, S M Taslim Uddin Raju, M M A Hashem. “Blood Donor Arrival Forecasting Using Regression Model and Analysis of Donor Behavioural Pattern.” *25th International Conference on Computer and Information Technology (ICCIT)*, pp. 897–902. IEEE, 2022. [doi:10.1109/ICCIT57492.2022.10054710](https://doi.org/10.1109/ICCIT57492.2022.10054710)
 - **Role**: reframed a 2.67%-base-rate classification task as aggregate regression over propensity buckets; built the decision-tree forecaster and the donor-behaviour analysis; evaluated the OneBlood deployment.

RESEARCH EXPERIENCE

Ongoing

FaithAudit: Do explainable deepfake detectors cite the region that caused their verdict? | PyTorch · [Code](#) 2026 – present

- An interventional audit of whether the facial region a detector cites drives its verdict, using each fake frame's frame-aligned authentic twin as an artifact-free counterfactual. Codebase complete and self-tested; first FaceForensics++ runs scheduled.

PredictorProbe: Is the I-JEPA predictor spatially grounded? | PyTorch · [Code](#) 2026 – present

- A label-free probe of whether the I-JEPA predictor uses the position it is queried at, ranking its output against every patch's true representation. Designed the protocol and built the codebase; first real-weights run complete; the undergraduates I supervise run it as their Final Year Design Project.

Earlier

Adversarial Robustness in Low-Resource NLP (Bangla) | PyTorch · [Code](#) 2025

- Character-, word- and noise-level attacks on Bangla language models, measuring which perturbations break which encoders; the groundwork for the Bangla NER paper above.

Multiclass SQL Injection Detection with a 1D CNN–TinyBERT Model | PyTorch · [Code](#) 2024 – 2025

Lightweight LLM and Reinforcement Learning for Text-to-SQL Generation | PyTorch · [Code](#) 2023

- TinyBERT text-to-SQL on Spider; RL from execution feedback raised logical correctness without increasing model size.

Undergraduate Thesis: BCI-Oriented EEG Signal Analysis and Classification | TensorFlow 2022 – 2023

- Cross-subject ERP detection on BCI Competition data: PSD/WPD/STFT features with NCA and Relief-F selection; the final CNN reached 92% cross-subject accuracy, most of the gain attributable to preprocessing and selection rather than the classifier.

TEACHING AND PROFESSIONAL EXPERIENCE

Lecturer, Department of CSE, Southeast University, Dhaka Feb 2024 – present

- Teach undergraduate programming, numerical methods and introductory artificial intelligence.
- Supervise six undergraduates on Final Year Design Projects in machine learning and NLP, including PredictorProbe (above).

Lecturer, Department of CSE, Uttara University, Dhaka Jul 2023 – Feb 2024

Part-time Lecturer, Department of CSE, Central University of Science and Technology, Dhaka Mar 2023 – Jul 2023

TECHNICAL SKILLS

Languages: Python, C++, Java. **ML / DL:** PyTorch, TensorFlow, Hugging Face, NumPy, SciPy, scikit-learn, Pandas, OpenCV.

Tools: Git, LaTeX, Kaggle.

HONORS AND AWARDS

- Dean's Award, Faculty of Electrical and Electronic Engineering, KUET, 2018–19 and 2019–20.
- Ranked 10th globally and 1st among Bangladeshi teams, Indian Rover Design Challenge (IRDC) 2020, Mars Society South Asia, as a member of Team Durbar, KUET.

SERVICE AND LEADERSHIP

- **Vice President, Hardware Acceleration Club of KUET (HACK)**, Mar 2022 – Mar 2023 — ran the microcontroller workshops and events.
- **Programmer, KUET Mars Rover (Team Durbar)**, Mar 2019 – Dec 2022 — rover navigation and on-board vision.

References: available upon request.